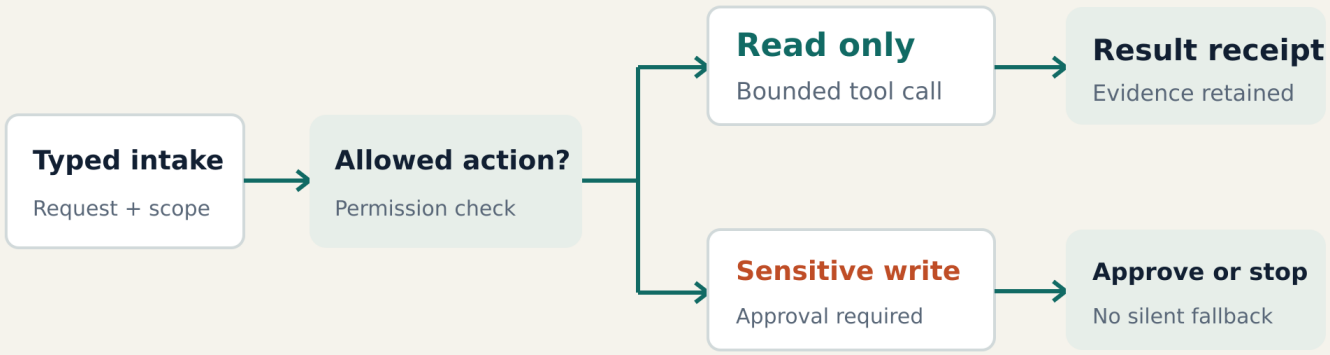
AI Task Routing and Human Approval Workflow

Define what can run, what needs approval, and what gets recorded.

AI SYSTEMS DESIGN Architecture sample

BOUNDED AUTOMATION Architecture sample



Failure stays visible; a proposed action is not a completed action.

THE PROBLEM

A useful automation needs to know when it is allowed to act. Otherwise a simple request can become an unreviewed record change, repeated action, or result nobody can explain.

HOW THE SAMPLE WORKS

Intake records the request and allowed scope. Read-only tasks may proceed through a bounded tool call. Sensitive writes wait for approval. Every route ends with an outcome or a visible exception.

WHAT YOU CAN INSPECT

The included action contract names an input reference, allowed action, approval state, idempotency key and result reference. Timeout, invalid input and denied approval each have a defined stop condition.

A PRACTICAL FIRST PROJECT

Begin with one read-only reporting workflow. Add a write step only after the approval rule, audit record and recovery behavior have been agreed.

A public architecture example. No production reliability benchmark, live connector execution or proprietary AES implementation is represented.

Action Contract and Failure Handling

A small public specification for a read-only exception report. The record below describes a demonstration request; it is not a receipt from a live system.

EXAMPLE ACTION CONTRACT

Field	Example value	Purpose
request_id	DEMO-001	Identify the task without exposing a user or customer.
input_ref	fictional_room_snapshot	Identify the approved input record.
allowed_action	read_only_report	Exclude changes to business records.
approval_state	not_required_for_read	Record why this read can proceed.
idempotency_key	demo_snapshot_v1	Prevent accidental duplicate work.
result_ref	null until produced	Never substitute a planned result for an observed one.
status	proposed	Change only after the action outcome is observed.

STOP CONDITIONS

Condition	Required response
Invalid input	Return a validation error and leave source records unchanged.
Permission denied	Stop and identify the missing authority; do not choose another write path.
Tool timeout	Return an exception with retry eligibility; do not report success.
Sensitive write requested	Require an explicit approval record before execution.

A design contract, not a security certification. A production implementation needs authentication, concurrency controls, audit storage and recovery tests.